

Our monthly volume forecasting models for large and small cap stocks are:

$$\begin{aligned} \text{Large Cap: } \Delta V(t) = & 0.0056 - 0.2990 \cdot \Delta V(t-1) + 0.1354 \cdot \Delta V(t-12) \\ & + 0.5461 \cdot \Delta \sigma_{\text{large}}(t) - 0.1877 \cdot \Delta \text{Sp}x \end{aligned} \quad (7.53)$$

$$\begin{aligned} \text{Small Cap: } \Delta V(t) = & 0.0029 - 0.2212 \cdot \Delta V(t-1) + 0.2385 \cdot \Delta V(t-12) \\ & + 0.3569 \cdot \Delta \sigma_{\text{small}}(t) + 0.3437 \cdot \Delta \text{Sp}x \end{aligned} \quad (7.54)$$

Main Observations:

- Monthly volumes exhibit trend reversion. The sign of the $\Delta V(t-12)$ variable was negative across both large and small cap stocks in each of the scenarios. If volume levels were up in one month, they were more likely to be down in the following month. If volume levels were down in one month, they were more likely to be up in the following month. This relationship is more significant for large cap than small cap stocks. Monthly volumes exhibit a positive seasonal trend. The sign of the $\Delta V(t-12)$ variable was positive and significant in most cases indicating a monthly pattern exists, although monthly volume levels vary. For example, December and August are consistently the lowest volume months during the year. October and January are the two highest volume months of the year (measured over our 10 year period). The relationship is stronger for small cap than for large cap stocks.
- Volumes are positively correlated with volatility. One way of thinking about this is that volume causes the volatility. Another explanation is that portfolio managers have a better opportunity to differentiate themselves and earn a higher return in times of increasing volatility. Hence they trade and rebalance their portfolios more often. The relationship here is slightly stronger for large cap stocks.
- The connection between volume and price level (SPX index) is the only factor that produces different relationships for large and small cap stocks. Small cap volume has always been positively related to price level. As the market increases, volume in small cap stocks increases, likely due to high investor sentiment during times of increasing market levels. Investors will put more in small cap stocks in a rising market hoping to earn higher returns, but will trade small stocks less often in a decreasing market. The strength of this relationship, has declined between the 2002–2006 and 2007–2011 periods. During 2002–2006 large cap stock volumes were also positively related to market price levels. But that relationship has reversed during 2007–2011.